

TOBY CRUST

XR ENGINEER — UNITY · AUGMENTED REALITY · REAL-TIME 3D

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Software engineer shipping **real-time 3D and AR** (Unity, Unreal Engine, C#, C++) to the App Store and Google Play, backed by **full-stack web** (React, TypeScript). Lead developer on an on-site AR heritage app for the NT Government, with AR and interactive work for Louis Vuitton's team, GeoArGames and a furniture-selection app before that.

SELECTED WORK

SLIK · NORTHERN TERRITORY GOVERNMENT

Augmented reality heritage app

Shipped a historical WWII AR app for the NT Government on iOS and Android, triggering on-site content via BLE beacons and geolocation. Built with Unity Addressables to reduce app store download size and enable modular, dynamic content updates for future expansion. Further locations and experiences are in production now, due to the success of the initial app.

Unity

C#

AR Foundation

ARKit / ARCore

BLE beacons

Addressables

Localization

Convai

SLIK · R&D

Holo Box: a flat screen that looks volumetric

Built a holographic display experience that tracks the viewer's head position with machine learning and renders the scene through an off-axis frustum, so the perspective updates to where the person actually is. The result reads as real depth inside the box rather than an image on a panel.

Unity

ML head tracking

Off-axis projection

Custom camera rig

PERSONAL PROJECT · UNREAL ENGINE 5

Movement shooter in UE5 and C++

Single-player movement shooter with a custom character movement component, a weapon system with recoil, reloads and hit markers, and AI bots that use perception, simulated aim and the same movement tech as the player. Includes a campaign mode with level progression and save games.

Unreal Engine 5

C++

Gameplay AI

Movement systems

Gameplay Tags

EXPERIENCE

Dec 2025 – present

Slik — Software Engineer, Sydney

- Lead developer on the NT Government AR app, from Unity build through to App Store Connect and Play Console release.
- R&D on Holo Box, an ML head-tracked holographic display.
- Full-stack client web work and AI agent workflows for internal tooling.

Oct 2024 – Oct 2025

Fabra — Software Engineer, Sydney

- Built features across Fabra's core SaaS product in React and TypeScript.
- Took over a departing senior engineer's workstream and shipped share-design permissions and email validation end to end, front and back.

Feb – Oct 2024

Bardon Design — Junior Software Engineer, Bangkok

- Built an interactive AR furniture selection app.
- Worked with the Louis Vuitton team on a gamified e-commerce experience built to lift engagement through interactive shopping.

Summers, 2021 – 2023

Software internships — Auckland

- Built an AR Rube Goldberg machine level for GeoArGames, and a 2D arcade game embedded in the Yuri&Niels site.

SIDE PROJECTS

Hexapod robot — real-time control

Six-legged walking robot. A three.js editor of the leg lets you drag joints or an IK foot handle and streams servo angles live over WiFi to Arduino firmware.

three.js

Inverse kinematics

C++

Python

TickyTick — cross-platform app

Local-first task manager for web, PWA and Android from one React codebase, with realtime Supabase sync and a Gemini function-calling assistant.

React

TypeScript

Capacitor

Supabase

TOOLKIT

ENGINES

Unity
Unreal Engine 5
three.js
Unity Addressables

AR & DEVICES

AR Foundation
ARKit / ARCore
BLE Beacons
Geolocation
Convai

LANGUAGES

C#
C++
TypeScript / JavaScript
Python

SHIP & OPERATE

App Store Connect
Google Play Console
Git / Unity Version Control
GitHub Actions

EDUCATION

Bachelor of Design Innovation — Victoria University, 2021–2023

OUTSIDE WORK

Robotics · 3D printing · Snowboarding · Waterpolo · Toastmasters

REFERENCES

Available on request — previous engineering leads at Fabra and Bardon Design are happy to speak.